

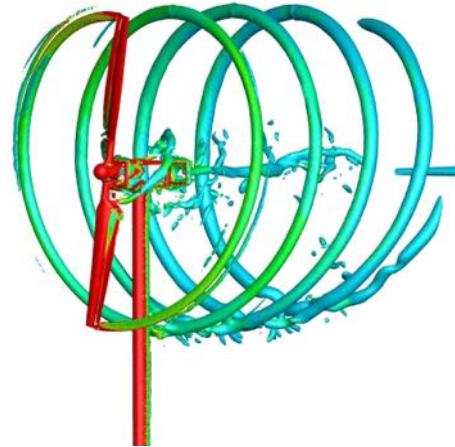
ENAE464 (CFD Lab) – Lecture



CFD Lab Lecture - 1

- **Quick Overview of CFD**
 - Fluid Dynamics
 - What is CFD?
 - CFD Analysis Process
 - Applications of CFD
 - What are pacing items?
 - What are some overall issues?
 - Simulation requirements
- **Numerical Lab**
 - Space Marching Quasi-1D
 - Unsteady Quasi-1D Euler
 - Sample Case of Nozzle

Wind Turbine in Shear Flow



I apologize for missing lecture time on Monday!

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ENAE 464 - CFD Lab

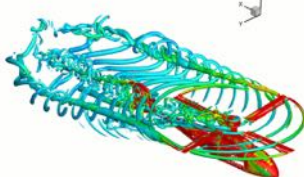


CFD Lab Lecture - 2

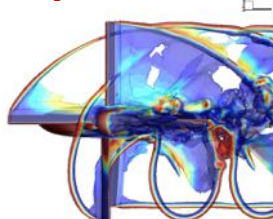
- **Dr. James D. Baeder**
 - E-mail: baeder@umd.edu
 - Phone: 301-405-1107 (3188 Martin Hall)
- Teach ENAE684 – CFD I every fall
- Sikorsky Professor
- Research – CFD, Rotorcraft, Wind Energy, Machine Learning and Quantum Computing



Main Rotor, Fuselage with Hub and Tail Rotor



Wing Lift-offset Rotor



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Fluid Dynamics



CFD Lab Lecture - 3

- **How can we understand phenomena associated with liquids and/or gases?**

Every so often, just the right combination of conditions and events occur to create an unbelievable event -- in this case an F/A-18 Hornet passing through the sound barrier. Not only were the water vapor, density and temperature just right, but there just happened to be a camera in the vicinity to capture the moment. Navy Ensign John Gay made this phenomenal photograph on July 7, 1999 while aboard the carrier USS Constellation as Navy Lt. Ron Candiloro flew by. The plane is actually in transonic flight, with normal shock waves emanating from behind the canopy and across the wings and fuselage.



<http://www.kettering.edu/~drussell/Demos/doppler/mach1.html>

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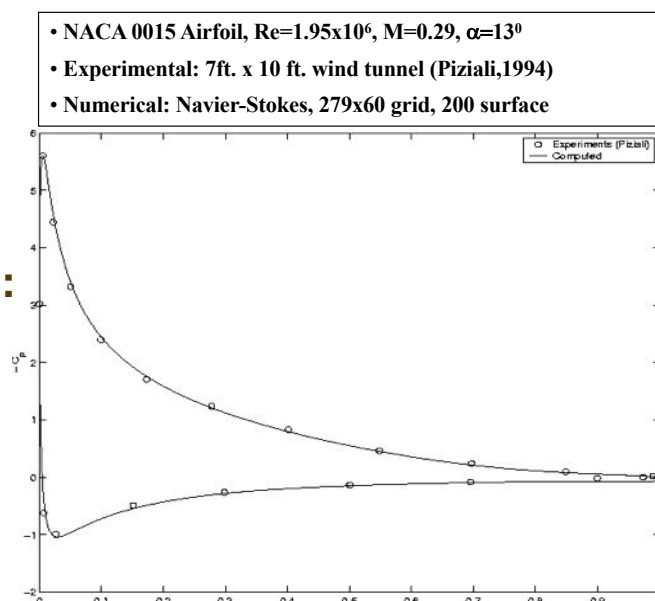
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Branches of Fluid Dynamics



CFD Lab Lecture - 4

- **Three Broad Categories**
 - **Experimental**
 - **Theoretical**
 - **Computational**
- **Crucial to know all three:**
 - **Each has their own advantages and disadvantages**
 - **Require verification and validation**



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Experimental Fluid Dynamics



CFD Lab Lecture - 5

- Only way to obtain some data
- Long set-up time
- Rapidly change parameters
 - E.g.: Angle of attack
- Often difficult to match all parameters
 - E.g.: Mach #, Reynolds #, Froude #, etc.
- Usually limited spatial and temporal resolution
- Need to establish experimental errors
- Repeatability issues?



<http://www.isvr.co.uk/graphics/bike5.jpg>

*Everyone believes the experimental results,
except the person who took the data!*

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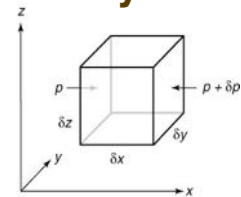
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Theoretical Fluid Dynamics



CFD Lab Lecture - 6

- Only way sometimes to see fundamental trends
 - May identify key non-dimensional parameters
- Fewer and fewer people trained in advanced theory
- Time consuming and tedious
- Usually significant simplifications required
 - E.g.: Linear theory, simple geometry, etc.
- Must ensure that simplifications are valid and that no mistakes are made in derivation



*Complicated planes (e.g. SR-71) developed w/o
computers and in record time... they understood
theoretical aerodynamics! (and had less red-tape)*

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Computational Fluid Dynamics



CFD Lab Lecture - 7

- **Fastest growing branch of fluid dynamics**
 - Computers getting cheaper and faster with more memory
 - Algorithms / grids improve to handle more complex flows
- **Can provide tremendous amounts of data**
 - Unsteady flow over complete aircraft
 - Can extract reduced order models for multi-disciplinary analysis
- **Need to understand/reduce numerics/assumptions**
 - Accurately predict simple problems, like laminar flow over flat plate?
 - Validation is key to confidence in results
- **CFD is not Color Fluid Dynamics**
 - Use of **color** should serve to better illustrate results



***Strong dichotomy about CFD results,
People automatically believe / disbelieve!***

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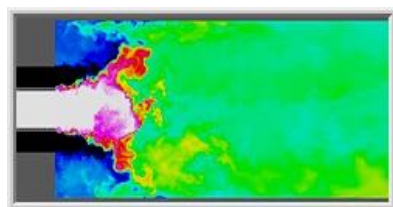
What is CFD?



CFD Lab Lecture - 8

- **CFD is:**

“The art and science of numerically solving partial differential equations at discrete points in space and time by replacing them with a set of algebraic equations that given the available computer resources best approximates the real geometry and fluid flow phenomena of interest.”



<http://ctr.stanford.edu/gallery/swirl.html>

Color shades of the instantaneous passive scalar in Large Eddy Simulations of turbulent flow in coaxial jet combustor with swirl. Simulations performed by Charles Pierce and Parviz Moin.

Let's break down the process!

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CFD Analysis Process



CFD Lab Lecture - 9

- **Problem Statement** --> information about the flow
- **Mathematical Model** --> IBVP = PDE + IC + BC
- **Mesh Generation** --> nodes/cells, time steps
- **Space Discretization** --> coupled ODE/DAE systems
- **Time Discretization** --> algebraic systems $Ax = b$
- **Iterative Solver** --> discrete function values
- **CFD Soft/Hardware** --> implementation, debugging
- **Simulation Run** --> parameters, stopping criteria
- **Postprocessing** --> visualization, analysis
- **Verification** --> model validation, adjustment

You can review this after class!

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Process: Problem Statement



CFD Lab Lecture - 10

- **What goals for the CFD analysis?**
 - Flow domain / phenomena to simulate?
 - Range of geometry / operating conditions?
- **What objectives for the CFD analysis?**
 - Integral quantities (e.g. lift, drag)?
 - Surface or field data (e.g. pressure or velocity vectors)?
 - Physical mechanisms (e.g. shock waves, separation)?
- **What type of flow (e.g. laminar / turbulent)?**
- **What is the easiest way to achieve the goal?**
 - Simplest / cheapest / fastest

CFD Analysis Process	
• Problem Statement	--> information about the flow
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One's goals and objectives should drive the simulation requirements!

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Problem Statement



CFD Lab Lecture - 11

- **Extract Engineering Data**

- Lift versus angle of attack
- Heat exchanged
- Wind tunnel correction effects
- Mass Flow
- Etc.

and/or

- **Extract Flow Physics**

- Mechanism for transition
- Separation / Vortex shedding
- Acoustic waves
- Aerodynamic interferences
- Dissociation
- Unsteady Shocks
- Etc.

Grid system, governing equations and model problem should be chosen to match simulation requirements!

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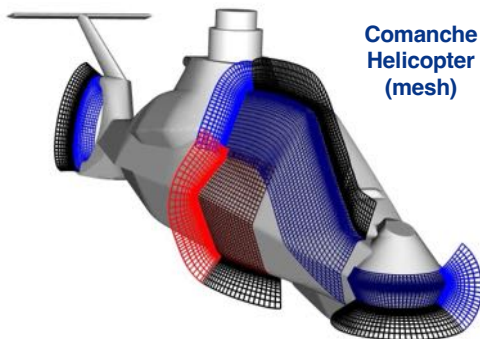
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CFD Applications

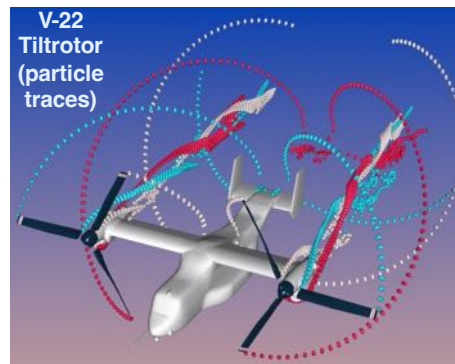


CFD Lab Lecture - 12

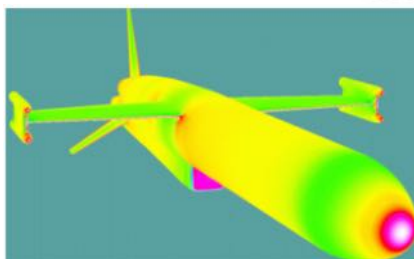
- **Aerodynamics**



Comanche
Helicopter
(mesh)

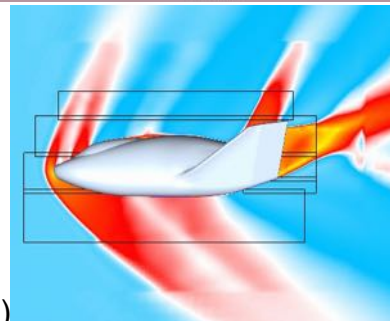


V-22
Tiltrotor
(particle
traces)



BQM-74
aerial target
(pressure
contours)

X-38
(Mach
contours)



Bob Meakin (retired)

http://rotorcrafterc.arc.nasa.gov/cfd/CFD4/New_Page/Home.htm

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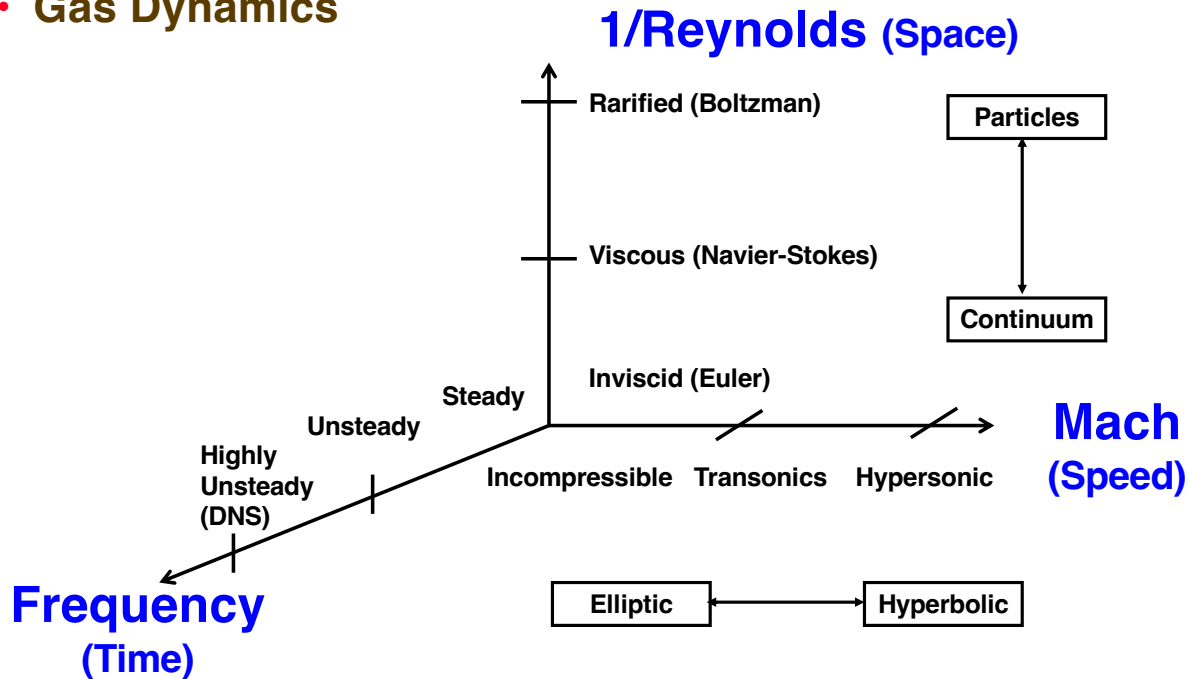
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Physics



CFD Lab Lecture - 13

• Gas Dynamics



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Process: Mathematical Model



CFD Lab Lecture - 14

- Choose suitable flow model and reference frame
- Define the computational domain
- Formulate conservation laws
- Determine constitutive relationships
- Specify initial / boundary conditions
 - *Just as important as interior governing equation*
- Simplify to reduce computational effort
 - Use symmetry and or reduced dimensions (1D / 2D)
 - Neglect unimportant terms if possible
 - Model small-scale fluctuations that cannot be captured

CFD Analysis Process	
• Problem Statement	--> information about the flow
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Keep it Simple! Now we have PDE / IC / BC

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Governing Equation: Physics



CFD Lab Lecture - 15

• Approximations based on Physics, Space and Time:

- Space Marching / 1D / 2D / Axisymmetric / 3D / Quasi-1D
- Linear / Small Disturbance / Non-linear
- Viscous / Inviscid / Boundary Layer
- Compressible / Incompressible
- Turbulent / Laminar
- Steady / Fully Unsteady (DNS) / LES / Reynold's Averaged
- Internal / External Flow
- Continuum / Non-Continuum
- Structural Coupling / Thermal Coupling / Ablation
- Physical Boundary Conditions
- General Purpose / Single Purpose
- Acoustics
- Chemistry
- Multiple fluids/phases
- Rotational / Non-rotational
- Etc.

Simplify!

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Conservation Laws - Reynold's Transport Theorem

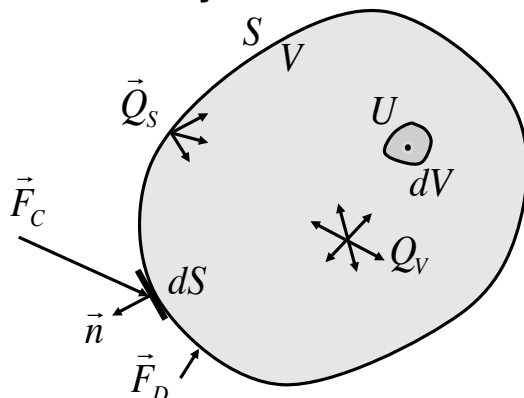


CFD Lab Lecture - 16

• Integral Form:

$$\frac{\partial}{\partial t} \int_V U dV = - \int_S (\vec{F} \cdot \vec{n}) dS + \int_S (\vec{Q}_s \cdot \vec{n}) dS + \int_V Q_v dV$$

• Arbitrary Control Volume:



U
 $\vec{v}$
 ρ



Osborne Reynolds
(an ordained priest)

$$\left(u \equiv U / \rho \quad \text{quantity / unit mass} \right)$$

• Discretize Directly in Physical Space

• Finite Volume Method

- McDonald, 1971; MacCormack & Paullay, 1972

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Process: Discretization



CFD Lab Lecture - 17

- **Mesh generation**
 - Structured / Unstructured • Single / Multiple meshes
- **Space discretization**
 - Finite differences / volumes / elements
- **Time discretization**
 - Explicit / implicit
- **Iterative solver**
 - Linearize and solve iteratively with Newton-like methods
 - Solve sequence of simpler linear subproblems

CFD Analysis Process	
• Problem Statement	--> Information about the flow
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Convert PDE system into large sparse algebraic system at each time step

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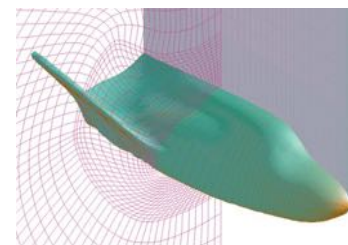
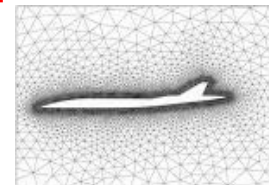
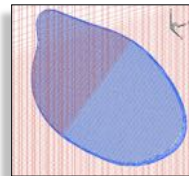
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Grid System (Mesh)



CFD Lab Lecture - 18

- **Topology**
 - Cartesian / Body-fitted
 - Structured / Unstructured
 - Single Mesh / Overset Meshes / Patched Meshes
- **Number of points**
 - Surface Resolution / Flowfield Resolution
 - Solution Adaptive
- **Moving / Stationary**



Grid system should be chosen to match governing equation and flow physics!

(Again, one would keep this as simple as possible for one's problem)

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Formulations



CFD Lab Lecture - 19

- **How does one derive the algebraic representations (discretization) to PDEs?**

- Finite Volume Methods
- Finite Difference Methods
- Finite Element Methods (FEM)
- Spectral Methods
- Boundary Element Methods (BEM)
- Particle Methods

$$\frac{\partial}{\partial t} \iiint Q dV + \iint F d\mathbf{A} = 0,$$

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} + \frac{\partial H}{\partial z} = 0$$

$$R_i = \iiint W_i Q dV^e$$

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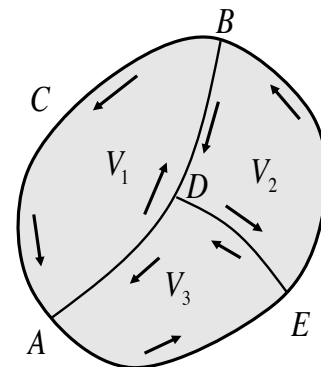
Finite Volume



CFD Lab Lecture - 20

- **If we assume that we have some average value in the volume (finite) and a flux at the sides one can write for each volume, i:**

$$\frac{\partial}{\partial t} U_i V_i + \sum_{j=1}^{\# \text{ of sides}} \vec{F}_{ij} \cdot \vec{S}_{ij} = \sum_{j=1}^{\# \text{ of sides}} \vec{Q}_{ij} \cdot \vec{S}_{ij} + Q_{V_i} V_i$$



- **N.B.:**
 - Easily extendible to arbitrary number of sides
 - Flux/surface source not defined at cell center or cell average
 - **All internal fluxes and surface sources cancel!**

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Differential Form



CFD Lab Lecture - 21

- A common approach is to convert integral to differential form using Gauss's theorem

$$\int_S (\vec{F} \cdot \vec{n}) dS = \int_V (\vec{\nabla} \cdot \vec{F}) dV$$

- For continuous fluxes and surface sources:

$$\int_V \frac{\partial U}{\partial t} dV + \int_V (\vec{\nabla} \cdot \vec{F}) dV = \int_V (\vec{\nabla} \cdot \vec{Q}_s) dV + \int_V Q_v dV$$

- Or, in divergence form:

$$\frac{\partial U}{\partial t} + \vec{\nabla} \cdot (\vec{F}_C - \vec{Q}_s + \vec{F}_D) = Q_v$$

- Can discretize using **finite differences** (Euler, 1768)

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Conservation Laws (2) Reynold's Transport Theorem

CFD Lab Lecture - 22



- General Differential Conservation Law Form
- Flux can take two forms:
 - Convective Flux (**w/ local fluid**)

$$\vec{F}_C \equiv \vec{v}U = \rho \vec{v}u \quad \left(u \equiv U/\rho \quad \text{quantity/unit mass} \right)$$

- Diffusive Flux (**away from concentration**)

$$\vec{F}_D \equiv -\chi \rho \vec{\nabla} u \quad (\chi \equiv \text{diffusivity constant})$$

- Then, one can write:

$$\frac{\partial \rho u}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}u - \chi \rho \vec{\nabla} u - \vec{Q}_s) = Q_v$$

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Navier-Stokes



CFD Lab Lecture - 23

- One can derive for N.S.:

$$\frac{\partial \rho u}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v} u - \chi \rho \vec{\nabla} u - \vec{Q}_s) = Q_v$$

- Mass (Continuity)

$$u \equiv 1; \quad \vec{Q}_s \equiv 0; \quad \chi \equiv 0; \quad Q_v \equiv 0$$

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = 0$$



- Similar for Momentum & Energy (see terms below)
- Can write as system of equations (vector notation):

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho \\ \rho \vec{v} \\ \rho E \\ \underbrace{\rho E}_U \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} \rho \vec{v} \\ \rho \vec{v} \otimes \vec{v} \\ \underbrace{\rho \vec{v} E}_{\vec{F}_C} \\ \underbrace{+pI - \bar{\tau}}_{-Q_s} \\ \underbrace{+\vec{v}p - \bar{\tau} \cdot \vec{v}}_{-Q_s} \\ \underbrace{-k\nabla T}_{\vec{F}_D} \end{pmatrix} = \begin{pmatrix} 0 \\ \rho \vec{f}_e \\ \underbrace{\rho \vec{f}_e \cdot \vec{v} + q_H}_{Q_v} \end{pmatrix}$$

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Euler Equations



CFD Lab Lecture - 24

- Let's assume
 - Inviscid
 - No external forces
 - 1-D
- Then --> 1-D Unsteady Euler Equations
 - Conservative Form



$$\frac{\partial \vec{Q}}{\partial t} + \frac{\partial \vec{F}}{\partial x} = 0$$

- Conservative solution vector and flux vector

$$\vec{Q} \equiv \begin{pmatrix} \rho \\ \rho u \\ e \equiv \rho E \end{pmatrix}; \quad \vec{F} \equiv \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (e + p)u \end{pmatrix}$$

- Where, for closure we need one additional (algebraic) equation

$$p = (\gamma - 1) \left[e - \frac{1}{2} \rho u^2 \right] \quad (\text{equation of state})$$

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1-D Unsteady Euler



CFD Lab Lecture - 25

- **Non-conservative Form (change of variables)**

$$\frac{\partial \vec{Q}}{\partial t} + [\hat{A}] \frac{\partial \vec{Q}}{\partial x} = 0$$

$$\frac{\partial \vec{Q}}{\partial \vec{Q}} \frac{\partial \vec{Q}}{\partial t} + \frac{\partial \vec{F}}{\partial \vec{Q}} \frac{\partial \vec{Q}}{\partial \vec{Q}} \frac{\partial \vec{Q}}{\partial x} = 0$$

- **Primitive solution vector and flux Jacobian matrix**

$$\vec{Q} \equiv \begin{pmatrix} \rho \\ u \\ p \end{pmatrix}; \quad [\hat{A}] \equiv \begin{pmatrix} u & \rho & 0 \\ 0 & u & 1/\rho \\ 0 & \rho c^2 & u \end{pmatrix}$$

$$[\hat{A}] = \frac{\partial \vec{Q}}{\partial \vec{Q}} \frac{\partial \vec{F}}{\partial \vec{Q}} \frac{\partial \vec{Q}}{\partial \vec{Q}}$$

- where

$$c^2 \equiv \frac{\gamma p}{\rho} \quad (\text{speed of sound})$$

*Eigenvalues are u, u+c and u-c
(flow velocity and acoustic waves)*

No guarantee of correct jump condition at discontinuity!

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Eigensystem



CFD Lab Lecture - 26

- **Investigate Eigensystem**

$$\det[\hat{A} - \lambda I] = (u - \lambda) \left[(u - \lambda)^2 - c^2 \right] = 0$$

- **Hyperbolic in time and nonlinear system (real eigenvalues)**

$$\lambda_1 = u, \quad \lambda_2 = u + c, \quad \lambda_3 = u - c$$

Note:

- Convection
- Acoustic Waves

- **Can decouple --> three scalar characteristic equations**

$$\frac{\partial \vec{W}}{\partial t} + [\Lambda] \frac{\partial \vec{W}}{\partial x} = 0$$

- where

$$\vec{W} = [X]^{-1} \vec{Q}; \quad [\Lambda] \equiv \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

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Characteristic Equations



CFD Lab Lecture - 27

- **Decoupled Characteristic Equations**
 - Simple scalar wave equations for characteristic quantities

$$\frac{\partial w_i}{\partial t} + \lambda_i \frac{\partial w_i}{\partial x} = 0 \quad \text{for } i=1,2,3$$

- Can solve in wave space and re-couple to physical space

Forms the basis of the method of characteristics

- **So, as a first step**
 - Can consider scalar PDEs (hyperbolic)!
 - Results can be extended to system of PDEs

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Simple Model Equations



CFD Lab Lecture - 28

- **Out of all the possible equations we will focus on the following in ENAE684:**

- **Time Accurate**

- Diffusion (Dissipation) Equation
- Convection (Advection) Equation
- Convection-Diffusion Equation

Physics ↔ Math

(parabolic)
(hyperbolic)
(parabolic)

- **Steady-State**

- Poisson's Equation

(elliptic)

Why focus on these?



Let's go through basic math and physics of each!

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Convection-Diffusion Eq.



CFD Lab Lecture - 29

Math

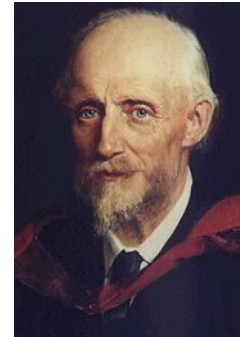
$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2} \quad (\text{parabolic}) \quad u_{xx}$$

• Solution

- Dependent on the **Mesh Reynold's Number**
 - Ratio of characteristic speed times length divided by diffusion

$$R_{\Delta} \equiv \frac{a \Delta x}{\nu}$$

- Similar to *physical* Reynold's number
- If
 - $R_{\Delta} \gg 2 \rightarrow$ convection dominated
 - $R_{\Delta} \ll 2 \rightarrow$ diffusion dominated
- Note:
 - Diffusion and phase shifting
 - If $a = u \rightarrow$ nonlinear (Burger's Equation)



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Convection-Diffusion Eq. (2)

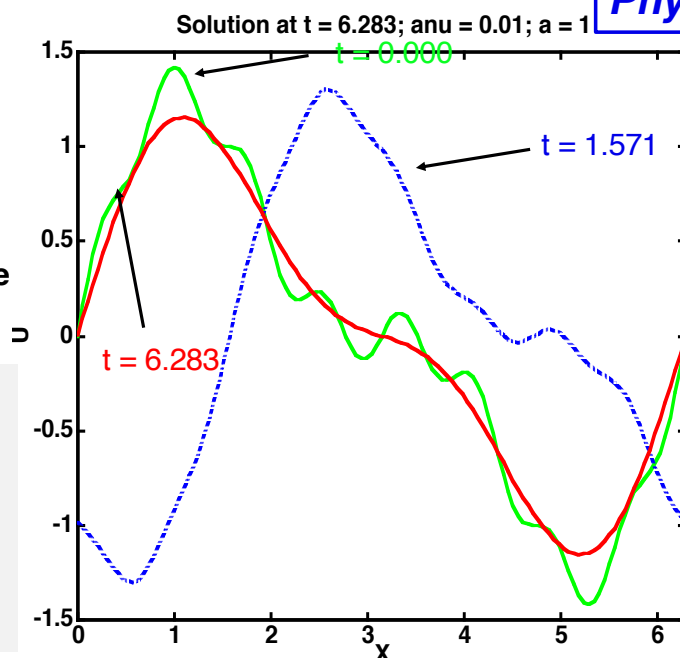
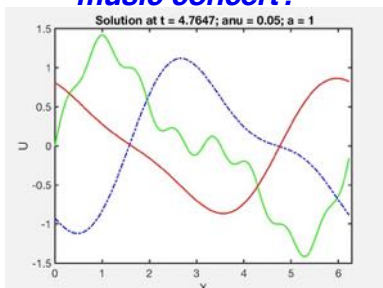


CFD Lab Lecture - 30

Physics

• Note:

- High Frequencies dampen out fastest
- Low frequencies dampen out slowest
- Note movement (convection) as well as damping with time
- **How is this like at a music concert?**



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Finite Volume / Difference



CFD Lab Lecture - 31

- **Finite Volume Methods** (McDonald, MacCormack & Paullay – 1970's)
 - Discretize *integral* form directly in the physical space
 - Convenient for making sure all internal fluxes cancel
 - Easy to extend to arbitrary unstructured meshes
 - Involves interpolation methods to derive flux interface values
 - Usually covered in most modern CFD courses
- **Finite Difference Methods** (Euler – 1768)
 - Based on definition from calculus
 - First applied to ordinary differential equations
 - Simplest to understand
 - Always covered in a CFD course



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So, We Convert PDE(s) to Algebraic System



CFD Lab Lecture - 32

- **PDE**
 - Continuous in time and space
 - Difficult to solve (analytical)
- **Set of ODE**
 - Continuous in time
 - Discrete in space
 - Useful (theoretically)
- **Algebraic Equations**
 - Discrete in both time & space
 - Linearize (if needed)
 - Easy to solve (numerical --> finite precision)

Example:

(Note: $u(x,t)$ can be scalar or vector)

$$\frac{\partial u}{\partial t} = - \frac{\partial (\text{flux}(u))}{\partial x} ; \quad \begin{array}{l} \text{boundary condition} \\ \text{initial condition} \end{array}$$



Discretize space derivative

$$\frac{d\vec{u}}{dt} = [A(\vec{u}, t)]\vec{u} - \vec{f}(t) = \vec{F}(\vec{u}, t) ; \quad \text{initial condition}$$



Discretize time derivative

$$[B]\vec{u}^{n+1} = \vec{F}(\vec{u}^n, \vec{u}^{n-1}) ; \quad \text{initial condition}$$

But, ... what about accuracy, stability, consistency, convergence?

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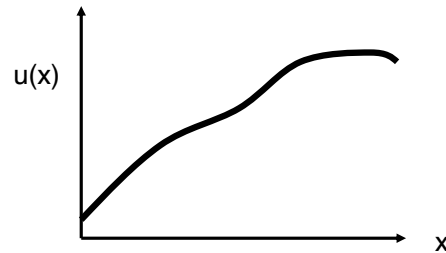
Analytical vs. Discrete



CFD Lab Lecture - 33

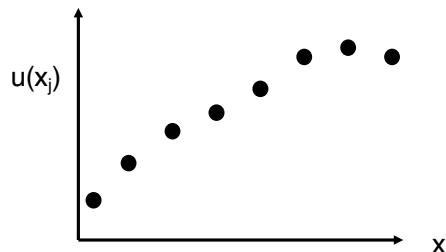
- Analytical**

- $u(x)$ known everywhere as function
- Exact representation of PDE
- Difficult to solve on computer
- Functional form may yield insight



- Discrete**

- $u(x_i)$ known only at discrete pts.
 - Possibility of aliasing (high freq. $\rightarrow$ low)
- Finite precision representation
 - Truncation $\rightarrow$ round-off error
- Suited to digital computer
- More difficult to gain insight



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Accuracy



CFD Lab Lecture - 34

- Differential:**

- **Unique**, since:

$$u_x \equiv \frac{\partial u}{\partial x} \equiv \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x}$$

- Difference:**

- **Not unique**, depends upon approximation and Δx :

$$u_x|_j \equiv f\left(\dots, u_{j-2}, u_{j-1}, u_j, u_{j+1}, u_{j+2}, \dots, u_x|_{j-2}, u_x|_{j-1}, u_x|_{j+1}, u_x|_{j+2}, \dots, \Delta x\right)$$

How does one determine the coefficients and the accuracy of difference approximations?

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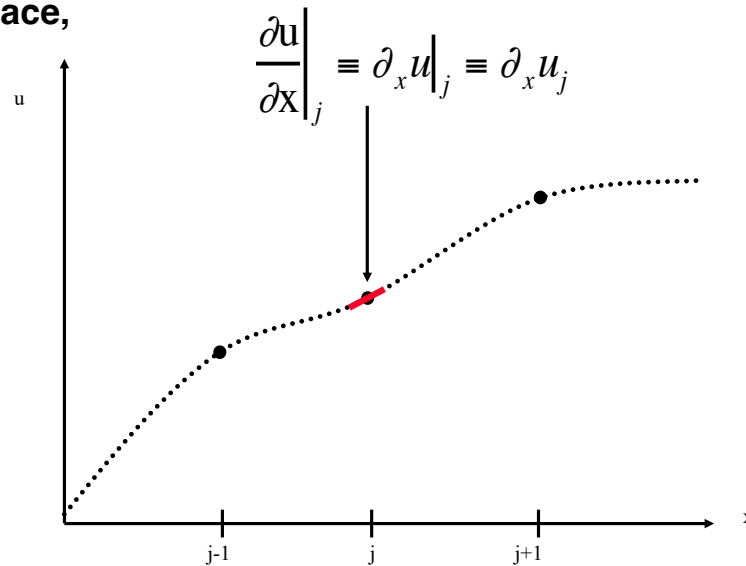
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Point Differential Operator

CFD Lab Lecture - 35

- Define relationship at one point in the mesh

- Eg. Space,



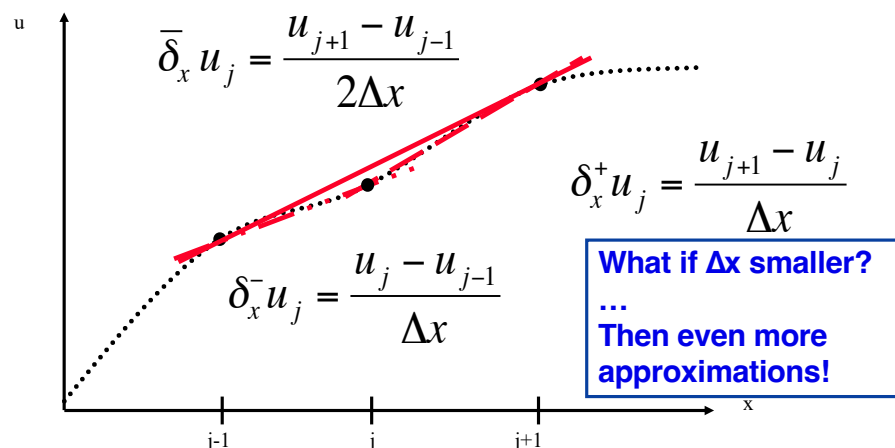
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Point Difference Operator

CFD Lab Lecture - 36

- Define relationship at one point in the mesh
 - Infinite number of difference operators available to approximate same differential operator (3 shown)!



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Taylor Series Analysis



CFD Lab Lecture - 37

- **From Calculus**

Taylor's Methodus Incrementorum Directa et Inversa (1715)



- If smooth, solution anywhere described by Taylor series expansion of function & derivatives at one point!:

$$u_{j+k} = u_j + (k\Delta x) \left(\frac{\partial u}{\partial x} \right)_j + \frac{1}{2} (k\Delta x)^2 \left(\frac{\partial^2 u}{\partial x^2} \right)_j + \dots + \frac{1}{n!} (k\Delta x)^n \left(\frac{\partial^n u}{\partial x^n} \right)_j$$

- So, for example:

$$u_{j+1} = u_j + \Delta x \left(\frac{\partial u}{\partial x} \right)_j + \frac{1}{2} \Delta x^2 \left(\frac{\partial^2 u}{\partial x^2} \right)_j + \dots + \frac{1}{n!} \Delta x^n \left(\frac{\partial^n u}{\partial x^n} \right)_j$$

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Taylor Series Analysis (2)



CFD Lab Lecture - 38

- **What if?**

What if not smooth?

- Subtract u_j from both sides
- Divide by Δx :

$$\frac{u_{j+1} - u_j}{\Delta x} = \left(\frac{\partial u}{\partial x} \right)_j + \frac{1}{2} \Delta x \left(\frac{\partial^2 u}{\partial x^2} \right)_j + \dots + \frac{1}{n!} \Delta x^{n-1} \left(\frac{\partial^n u}{\partial x^n} \right)_j$$

- This is a first order approximation to the first partial derivative in space of u at point j

This process can be formalized

- ***Taylor Tables to reduce chances for errors***
(Method of Undetermined Coefficients)

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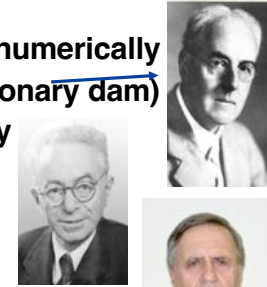
Numerical Methods



CFD Lab Lecture - 39

• A Brief *Early History of CFD*

- Pre 1900 - Euler and Runge-Kutta methods for solving ODEs numerically
- 1910 - [Richardson](#) iteratively solved PDE - Laplace's eq. (masonry dam)
- 1928 - Definition of CFL number by [Courant](#), Friedrichs & Lewy
- 1950 - von Neumann method for stability analysis
- 1950 - Frankel applied Successive Over Relaxation (SOR)
- 1950 - Artificial viscosity by von Neumann and Richtmeyer
- 1955 - Alternating Direction Implicit (ADI) by Peaceman and Rachford
- 1957 - Particle-in-cell (PIC) by Evans and Harlow
- 1959 - [Godunov](#) develops Riemann-based scheme for compressible flow
- 1960 - [Lax-Wendroff](#) scheme for second order accurate shock calculations
- 1962 - Multigrid developed by Fedorenko
- 1966 - Shock Fitting methods developed
- 1969 - MacCormack's practical 2-step Lax-Wendroff variant
- 1971 - [Murman](#) and Cole develop scheme for handling TSD
- 1973 - Boris and Book develop Flux Correction Transport (FCT) methods



Advances Made by Mathematicians, Physicists and Engineers

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Process: CFD Simulations



CFD Lab Lecture - 40

- **Computing times for simulation depend on**
 - Choice of mathematical model
 - Choice of numerical algorithms and data structures
 - Programming language (usually Fortran for CFD codes)
 - Software implementation
 - Linear algebra tools, stopping criteria for iterative solvers
 - Discretization parameters (mesh size, time step)
- **Run simulation for given parameters**
 - Hardware (memory, cache, I/O, CPU, GPU)
 - Vectorization and Parallelization
 - Workstation, Cluster, or HPC Nodes

CFD Analysis Process

• Problem Statement	-> information about the flow
• Mathematical Model	-> IBVP = PDE + IC + BC
• Mesh Generation	-> nodes/cells, time steps
• Space Discretization	-> coupled ODE/DAE systems
• Time Discretization	-> algebraic systems $Ax = b$
• Iterative Solver	-> discrete function values
• CFD Soft/Hardware	-> implementation, debugging
• Simulation Run	-> parameters, stopping criteria
• Postprocessing	-> visualization, analysis
• Verification	-> model validation, adjustment

Quality of simulation depends on these factors!

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Computers

CFD Lab Lecture - 41

- **Early ‘Computational’ methods developed before digital computers**
 - Calculations done by hand, often in teams!
- **Mechanical/Analog Computers in 1930’s**
- **Digital computers available in 1950’s**
 - First 6 computer programmers were women (1946)
- **Computer ‘cost’ drops 6 orders of magnitude in 20 yrs.**
 - PC’s today have power of supercomputer of 10 years ago
 - Clusters of 10,000’s of PC’s forming supercomputer of today
 - GPGPU’s? Smart phones?



Pickering's Women



Differential Analyzer



Electronic Numerical Integrator And Computer

**CFD homework assignments today --->
Could have been PhD thesis in the 1950’s or 1960’s**

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Process: Postprocessing

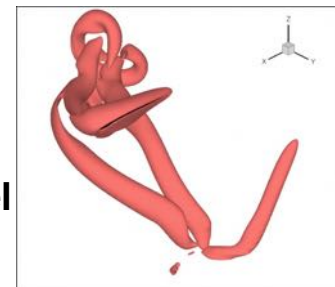
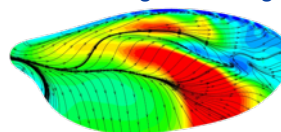
CFD Lab Lecture - 42

- **Extract desired information from computed flow field**
 - **Derived quantities (skin friction, vorticity)**
 - **Integral parameters (lift, drag)**
 - **Statistical data (turbulence)**
 - **Visualization as images**
 - 1-D symbols and lines
 - 2-D contours and surfaces
 - 3-D cutplanes, isosurfaces
 - Vectors, particle traces, animations
 - **Use to debug, verify and validate CFD model**
 - **Movies**

CFD Analysis Process

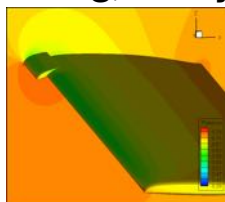
Problem Statement	-> information about the flow
Mathematical Model	-> IBVP = PDE + IC + BC
Mesh Generation	-> nodes/cells, time steps
Space Discretization	-> coupled ODE/DAE systems
Time Discretization	-> algebraic systems $Ax = b$
Iterative Solver	-> discrete function values
CFD Soft/Hardware	-> implementation, debugging
Simulation Run	-> parameters, stopping criteria
Postprocessing	-> visualization, analysis
Verification	-> model validation, adjustment

Rotating Insect Wing



Insect Flapping

Moving Slat



Don't skimp on analysis!

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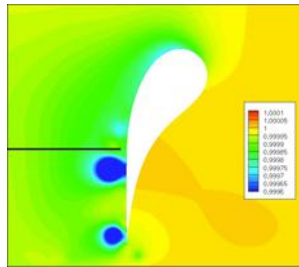
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CFD Applications (2)

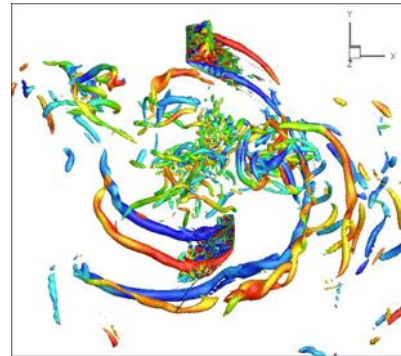


CFD Lab Lecture - 43

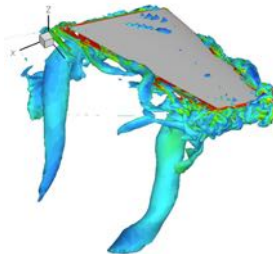
- **Shrouded rotor**



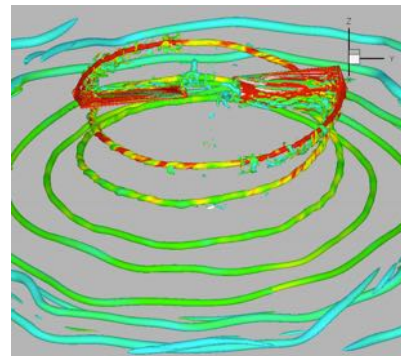
- **Cyclorotor**



- **Avian Flapping**



- **Rotor in Ground Effect**



Some of my applications!

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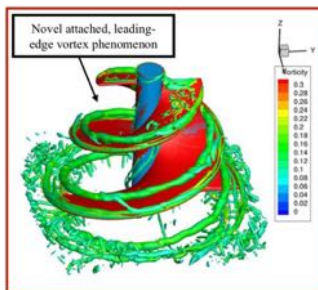
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CFD Applications (3)

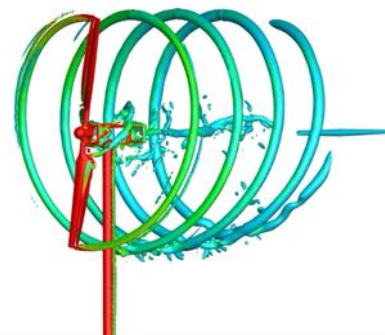


CFD Lab Lecture - 44

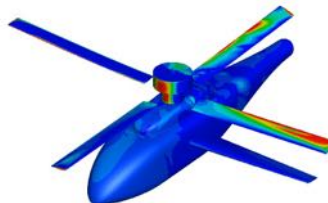
- **DaVinci Rotor**



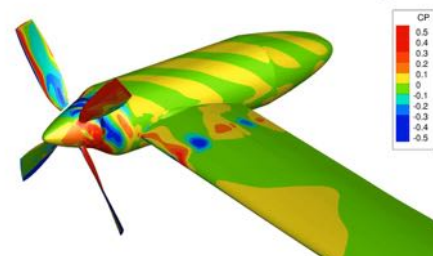
- **Wind Turbine with Wind Shear**



- **Wing Lift-Offset Rotorcraft**



- **Propeller/ Wing Interaction**



Some of my applications!

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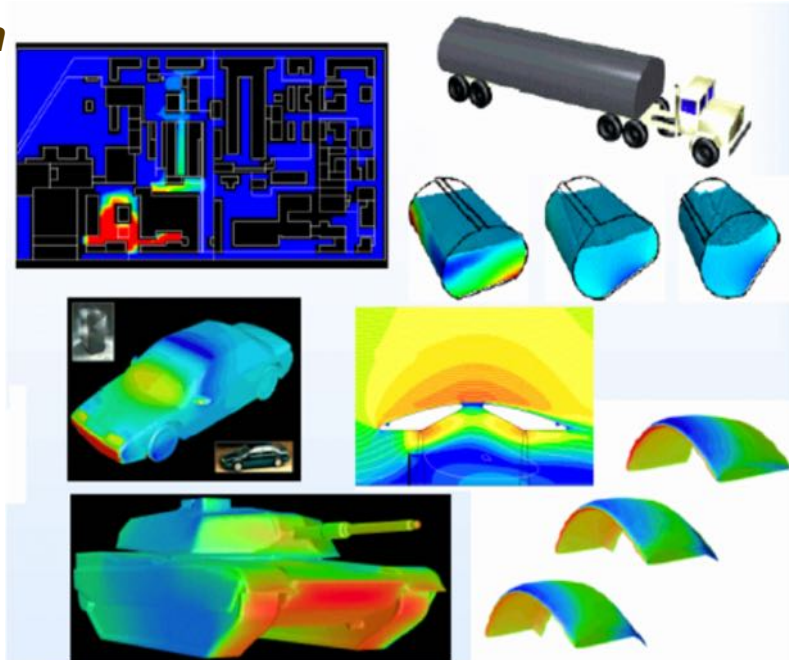
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CFD Applications (4)



CFD Lab Lecture - 45

- **Weather Prediction**
- **Wakes**
- **Automobiles**
- **Wind Loading**
- **Environmental**
- **Turbulence**
- **Galaxy Evolution**
- **Tidepools**



More than just Aerospace!

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Process: Verification



CFD Lab Lecture - 46

- **Build *trust* in the CFD solutions**
 - **Acknowledged errors**
 - Uncertainty in physical modeling (e.g. turbulence models)
 - Discretization errors
 - Iterative convergence errors
 - Round-off errors (finite precision)
 - **Unacknowledged errors**
 - Computer programming errors (bugs)
 - Usage errors (wrong parameters, models or bc)
- **Verify CFD codes to reduce both sources of errors**
 - Compare to known solutions – analytic or numerical

CFD Analysis Process	
• Problem Statement	--> information about the flow
• Mathematical Model	--> IBVP = PDE + IC + BC
• Mesh Generation	--> nodes/cells, time steps
• Space Discretization	--> coupled ODE/DAE systems
• Time Discretization	--> algebraic systems $Ax = b$
• Iterative Solver	--> discrete function values
• CFD Soft/Hardware	--> implementation, debugging
• Simulation Run	--> parameters, stopping criteria
• Postprocessing	--> visualization, analysis
• Verification	--> model validation, adjustment

***Finally can validate with experiment,
run new cases when have trust in CFD!***

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What is CFD? (2)



CFD Lab Lecture - 47

In Other Words, to Summarize:

- **CFD is an engineering and scientific tool:**

- Similar to theoretical or experimental tools

- That, based on:

- Geometry
- Fluid flow physics
- Computer

- One selects the appropriate:

- Governing equations
- Numerical method
- Grid system

- Solves using CFD codes

- And then examine the solutions:

- To finally obtain greater physical insight and/or engineering results!

CFD Analysis Process

- | | |
|------------------------|-----------------------------------|
| • Problem Statement | --> information about the flow |
| • Mathematical Model | --> IBVP = PDE + IC + BC |
| • Mesh Generation | --> nodes/cells, time steps |
| • Space Discretization | --> coupled ODE/DAE systems |
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Pacing Items in CFD



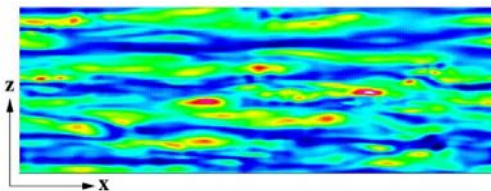
CFD Lab Lecture - 48

- **Computers**



RC is a cluster of dual-xenon processors at the University of Maryland that is used to calculate the aerodynamics, acoustics and structural response of rotorcraft.

- **Turbulence**



<http://ctr.stanford.edu/gallery/sf.html>

Color shades of the skin friction in the Direct Numerical Simulations of fully developed turbulent channel flow. The simulations were performed by John Kim, Parviz Moin and Robert Moser (Journal of Fluid Mechanics, v 177, pp 136-166, 1987). The plot is produced by Arthur Kravchenko.

- **Algorithms**

$$\phi_{xx} + \phi_{yy} = f(x, y) \longrightarrow \phi_{i,j}^{n+1} = \frac{1}{4}(\phi_{i+1,j}^n + \phi_{i-1,j}^n + \phi_{i,j+1}^n + \phi_{i,j-1}^n) + \Delta^2 f_{i,j}^n$$

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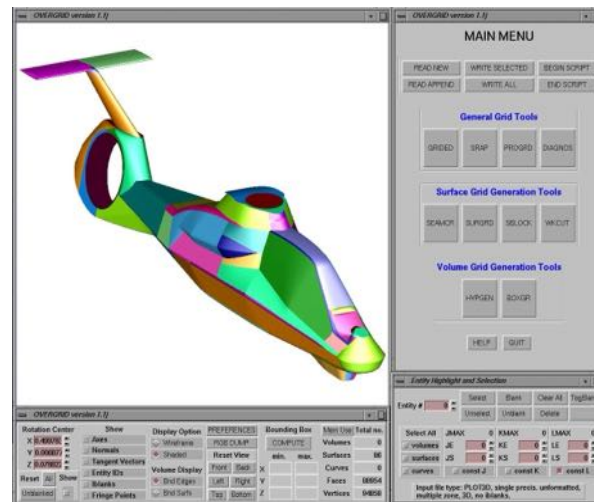
Pacing Items in CFD (2)



CFD Lab Lecture - 49

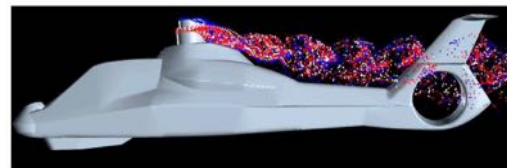
- **Geometry Definition**

- **Grids**



<http://rotorcraft.arc.nasa.gov/cfd/CFD4/CGT/overgrid.html>

- **Pre- and Post-Processing**



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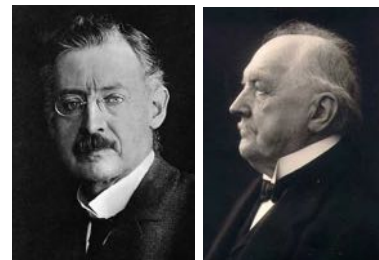
Numerical Issues



CFD Lab Lecture - 50

- **Accuracy, Stability, Consistency & Convergence**

- Separate time and space or combined?
- Upwind / Central and/or Compact for space?
- Low order / high order / adaptive stencil for space?
- Numerical boundary conditions?
- Implicit / Explicit in time?
- Runge-Kutta / Multi-step / Higher-Derivatives for time?
- How does the method dissipate or disperse the solution?
- Are their numerical limits on the time step size?
- How does method behave for discontinuous solutions?
- Is the method conservative?
- What about nonlinear systems / non-Cartesian meshes, etc... ?



These issues would be covered extensively in CFD

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Space Marching Quasi-1D Euler



CFD Lab Lecture - 51

- Consider steady **quasi-1D** Euler equations:

$$\frac{\partial \rho u A}{\partial x} = 0 \quad \text{Continuity Equation}$$

$$\frac{\partial (\rho u^2 + p) A}{\partial x} = p \frac{\partial A}{\partial x} \quad \text{X - Momentum Equation}$$

$$\frac{\partial (e + p) u A}{\partial x} = 0 \quad \text{Energy Equation}$$

**NOTE: Only
space
derivative!**

- Rewrite in non-conservative form:

$$uA \frac{\partial \rho}{\partial x} + \rho A \frac{\partial u}{\partial x} + \rho u \frac{\partial A}{\partial x} = 0$$

$$u^2 A \frac{\partial \rho}{\partial x} + 2\rho u A \frac{\partial u}{\partial x} + A \frac{\partial p}{\partial x} + (\rho u^2 + p) \frac{\partial A}{\partial x} = p \frac{\partial A}{\partial x}$$

$$\frac{1}{2} u^3 A \frac{\partial \rho}{\partial x} + \left[\frac{\gamma}{\gamma-1} p A + \frac{3}{2} \rho u^2 A \right] \frac{\partial u}{\partial x} + \frac{\gamma}{\gamma-1} u A \frac{\partial p}{\partial x} = - \left[\frac{\gamma}{\gamma-1} p u + \frac{3}{2} \rho u^3 \right] \frac{\partial A}{\partial x}$$

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Space Marching Quasi-1D Euler



CFD Lab Lecture - 52

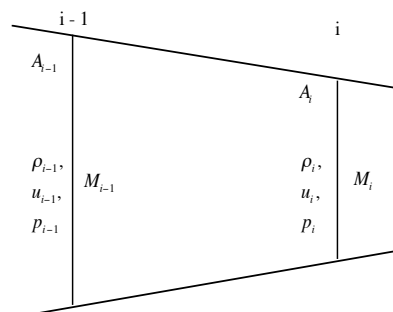
- Can write differential in matrix form:

$$\begin{bmatrix} u & \rho & 0 \\ u^2 & 2\rho u & 1 \\ \frac{1}{2}u^3 & \frac{\gamma}{\gamma-1}p + \frac{3}{2}\rho u^2 & \frac{\gamma}{\gamma-1}u \end{bmatrix} \begin{pmatrix} \delta \rho \\ \delta u \\ \delta p \end{pmatrix} = \begin{pmatrix} -\rho u \frac{\delta A}{A} \\ -\rho u^2 \frac{\delta A}{A} \\ -\left[\frac{\gamma}{\gamma-1} p u + \frac{1}{2} \rho u^3 \right] \frac{\delta A}{A} \end{pmatrix}$$

- Can we march from one station to the next in flow due to area change?

- After manipulation:

$$\delta \rho = \rho \frac{M^2}{1 - M^2} \frac{\delta A}{A}$$



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Space Marching Quasi-1D Euler

CFD Lab Lecture - 53



- Can discretize:

$$\rho_i = \rho_{i-1} + \rho \frac{M^2}{1-M^2} \frac{A_i - A_{i-1}}{A}$$

- From continuity of mass and energy:

$$u_i = \frac{\rho_{i-1} u_{i-1} A_{i-1}}{\rho_i A_i}; \quad p_i = \rho_i \left[\frac{\gamma-1}{2\gamma} (u_{i-1}^2 - u_i^2) + \frac{p_{i-1}}{\rho_{i-1}} \right]$$

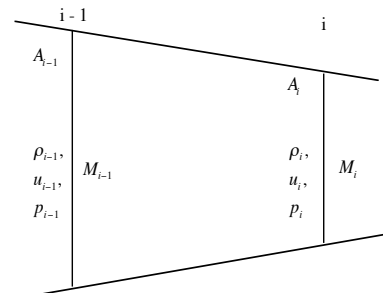
- For initial guess can freeze coefficient at old value:

$$\rho \frac{M^2}{1-M^2} \frac{1}{A} = \rho_{i-1} \frac{M_{i-1}^2}{1-M_{i-1}^2} \frac{1}{A_{i-1}} \Rightarrow \bar{\rho}_i, \bar{u}_i, \bar{p}_i \Rightarrow \bar{M}_i$$

- Can improve (2nd order):

$$\rho \frac{M^2}{1-M^2} \frac{1}{A} = \left(\frac{\bar{\rho}_i + \rho_{i-1}}{2} \right) \frac{\left(\frac{\bar{M}_i + M_{i-1}}{2} \right)^2}{1 - \left(\frac{\bar{M}_i + M_{i-1}}{2} \right)^2} \left(\frac{2}{\bar{A}_i + A_{i-1}} \right) \Rightarrow \bar{\bar{\rho}}_i, \bar{\bar{u}}_i, \bar{\bar{p}}_i \Rightarrow \bar{\bar{M}}_i$$

NOTE: Now algebraic equations!



Space Marching Quasi-1D Euler

CFD Lab Lecture - 54



- What if a shock?:

- Fit the shock:

$$\rho_2 = \rho_1 \left(1 + \frac{M_1^2 - 1}{\frac{\gamma-1}{2} M_1^2 + 1} \right)$$

- From continuity and energy

$$u_2 = \frac{\rho_1 u_1}{\rho_2}; \quad p_2 = \rho_2 \left[\frac{\gamma-1}{2\gamma} (u_1^2 - u_2^2) + \frac{p_1}{\rho_1} \right]$$

$$A_1 = A_2$$

$$\left. \begin{array}{l} \rho_1 \\ u_1 \\ p_1 \\ M_1 \end{array} \right| \left. \begin{array}{l} \rho_2 \\ u_2 \\ p_2 \\ M_2 \end{array} \right|$$

- May need to iterate to get correct shock location!

- Can use Newton iteration!

Quasi-1D Euler Review



CFD Lab Lecture - 55

- **Quasi-1D Euler**
 - **Space Marching**
 - Rewrite in non-conservative form
 - Put differential in matrix form
 - Solve for change in density
 - Use continuity of mass and energy to get velocity and pressure
 - Can iterate as march to get 2nd order
 - Can fit the shock
 - May need to iterate to get shock location for given back pressure
 - **Notes briefly discuss exact solution**
 - Use area for sonic flow!
 - Shock jump condition

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Sample Case in Nozzle

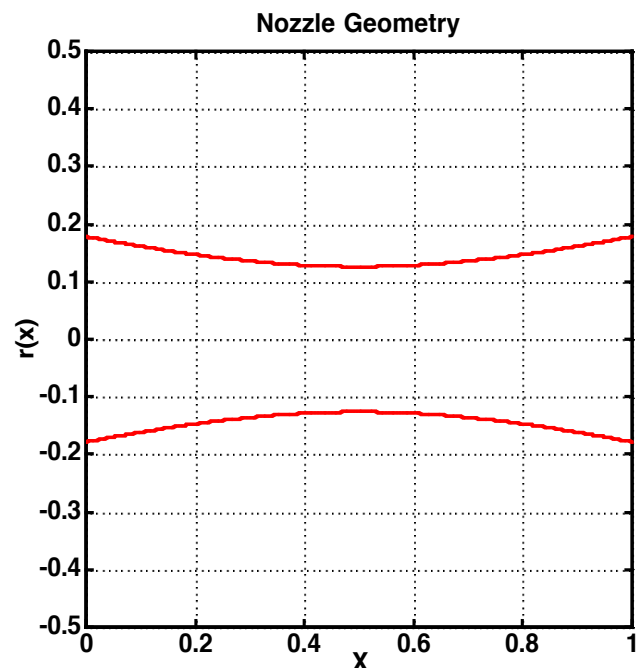


CFD Lab Lecture - 56

- **Area given by:**

$$A(x,t) = \frac{1}{20} \left[1 + 4 \left(x - \frac{1}{2} \right)^2 \right] \quad 0 \leq x \leq 1$$

- **Can plot the nozzle geometry:**



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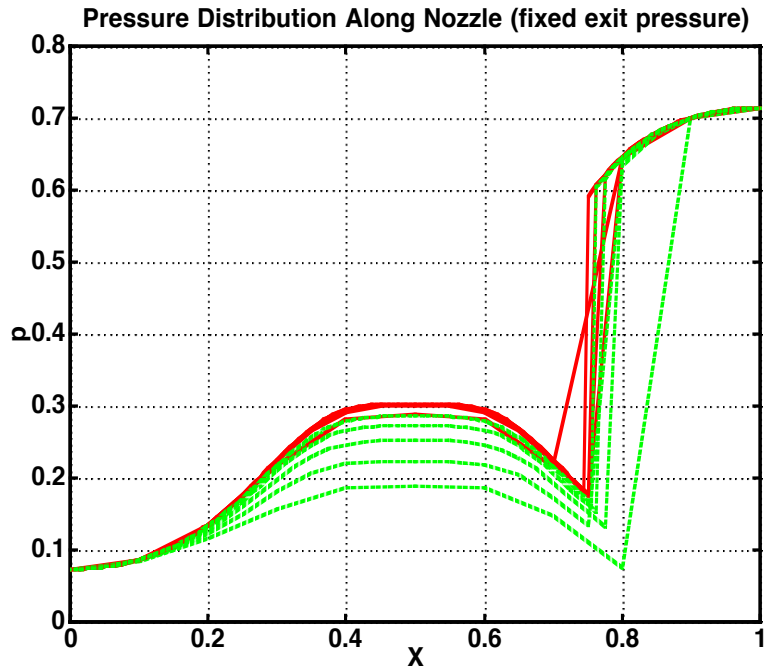
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Sample Case in Nozzle (2)



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- **How solution (space-marching) converges with mesh size**
 - Fixed back pressure
 - Iterates for shock location!



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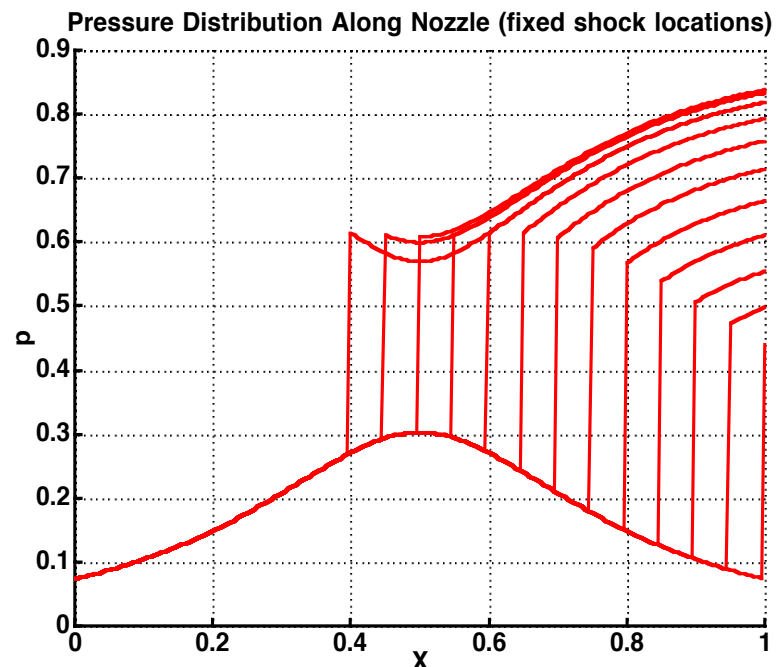
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Sample Case in Nozzle (3)



CFD Lab Lecture - 58

- **How solution (space-marching) varies with shock location**
 - Changes the back pressure



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Unsteady Quasi-1D Euler

CFD Lab Lecture - 59

- **Consider vector in form:**

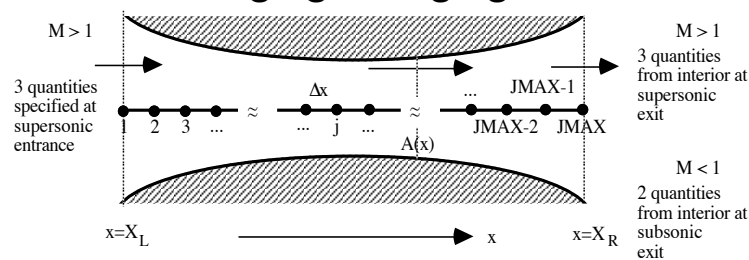
$$\frac{\partial \bar{Q}}{\partial t} + \frac{\partial \bar{F}}{\partial x} = \bar{S}$$

- **Where:**

$$\bar{Q} = \begin{pmatrix} \rho A \\ \rho u A \\ e A \end{pmatrix}, \quad \bar{F} = \begin{pmatrix} \rho u A \\ (\rho u^2 + p) A \\ (e + p) u A \end{pmatrix}, \quad \bar{S} = \begin{pmatrix} 0 \\ p \frac{\partial A}{\partial x} \\ 0 \end{pmatrix}$$

$$p = (\gamma - 1)(e - 0.5\rho u^2)$$

- **For a transonic converging/diverging nozzle:**



Unsteady Quasi-1D Euler (2)

CFD Lab Lecture - 60

- **In interior use flux-vector splitting:**
 - Too complicated to discuss details here

$$\Delta \hat{Q}_j^n = -\frac{\Delta t}{\Delta x} (\hat{F}_j^{+n} - \hat{F}_{j-1}^{+n} + \hat{F}_{j+1}^{-n} - \hat{F}_j^{-n}) + \frac{\Delta t}{\Delta x} \begin{pmatrix} 0 \\ p_j^n (A_{j+1/2}^n - A_{j-1/2}^n) \\ 0 \end{pmatrix}$$

$$RESID_i^n = \Delta \hat{Q}_i^n$$

$$\hat{Q}_j^{n+1} = RESID_j^n + \hat{Q}_j^n$$

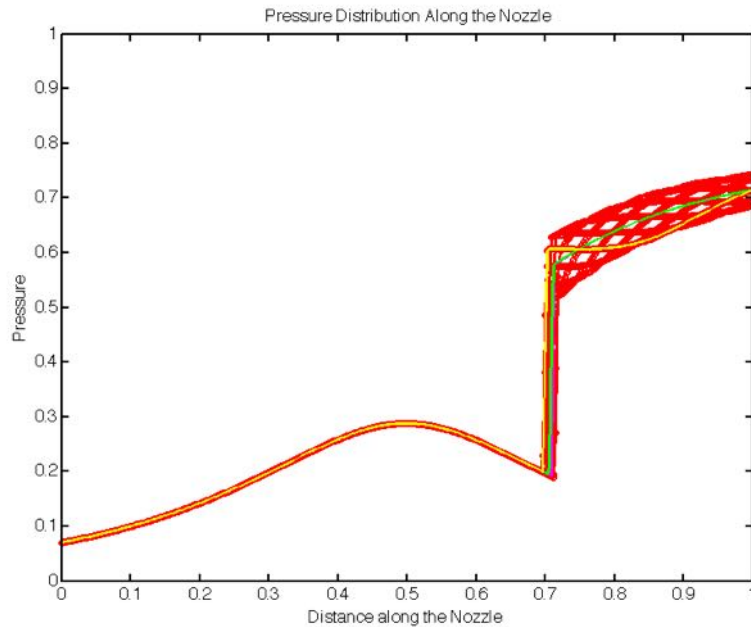
- **And apply appropriate bc's at ends**
- **Make sure appropriate area function**
- **Make sure appropriate initial condition with supersonic entrance!**

Sample Case in Nozzle (4)



CFD Lab Lecture - 61

- **Unsteady back pressure!**
 - Start with space marching
 - Then, run unsteady solver to w/ steady back pressure
 - Then, vary the back pressure!
 - See pressure waves move upstream



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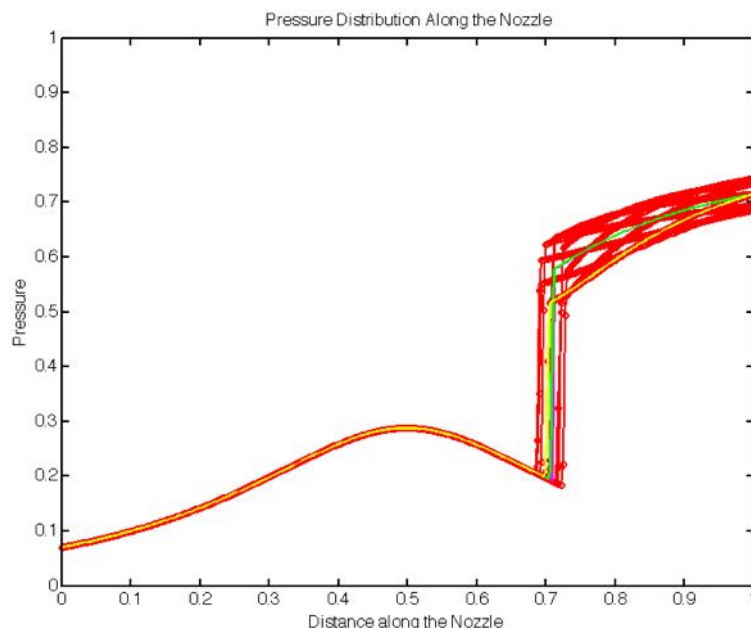
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Sample Case in Nozzle (5)



CFD Lab Lecture - 62

- **Unsteady back pressure!**
 - Have the back pressure change more slowly
 - Closer to quasi-steady oscillation, but still some time history effects



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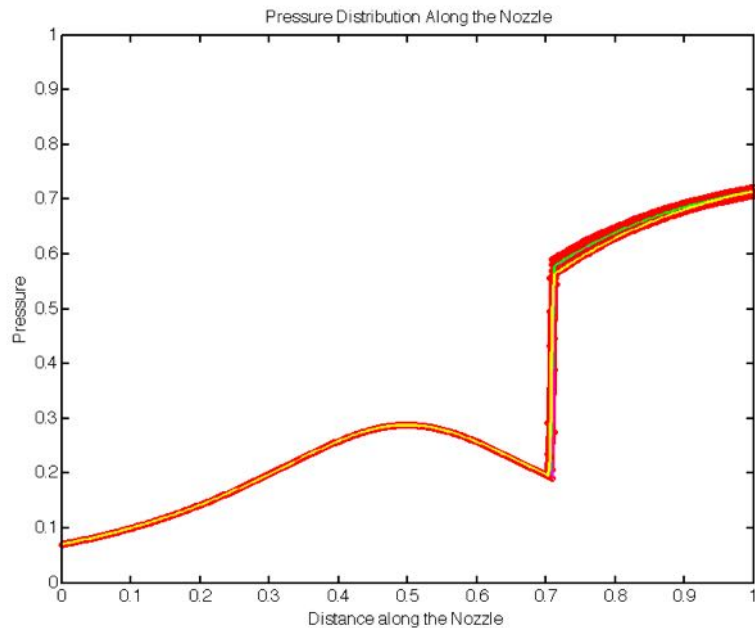
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Sample Case in Nozzle (6)



CFD Lab Lecture - 63

- **Unsteady back pressure!**
 - Now, a smaller change in back pressure



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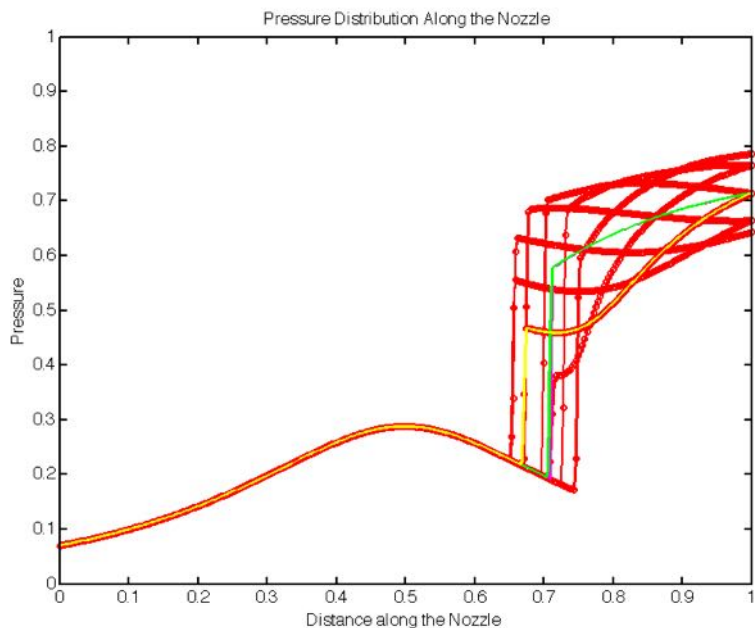
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Sample Case in Nozzle (7)



CFD Lab Lecture - 64

- **Unsteady back pressure!**
 - Now, a larger change in back pressure



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Sample Case in Nozzle (8)



CFD Lab Lecture - 65

- **Unsteady back pressure!**
 - Now, a very large change in back pressure
 - Nozzle actually unstarts
 - Shock passes upstream of the throat

